

Trained



train



Large labelled
training set

Corpus-specialized
features

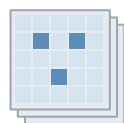
Standard



Cross-env.



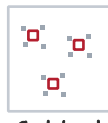
Training-free, prior work



Frozen patch
features

Standard

→



Codebook



Segmentation CLS attention

or

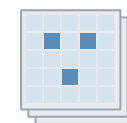


Prior-selected / aggregated features

Cross-env.

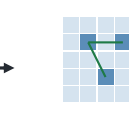


Ours

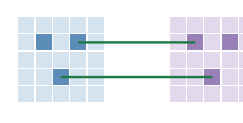


Frozen patch
features

Standard



Within-image
Balanced, matchable patch evidence



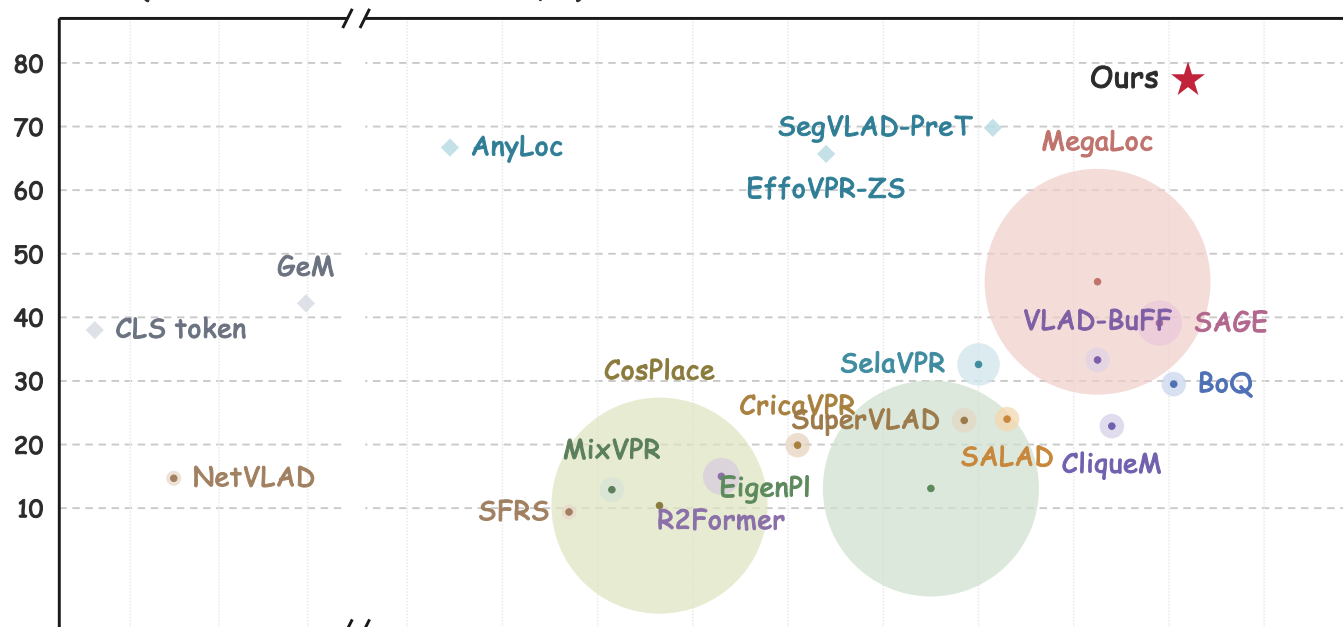
Across-image
Balanced, matchable patch evidence

Cross-env.



Supervision cost vs. cross-environment recall

VP-Air R@1 (cross-environment benchmark, %)



Tokyo 24/7 R@1 (standard benchmark, %)

★ ours (training-free)

♦ training-free

● trained

● 1 million images

● 5 million images

Recall across 17 benchmarks

